Matteo **Brucato**

Helping humans make better decisions.

Co-Founder & Chief Scientific Officer
OSM Data
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EDUCATION

Ph.D. in Computer Science
UMass Amherst
Amherst, MA, USA · 2013 – 2021
Thesis: Package Queries — Enabling Declarative and Scalable Prescriptive Analytics in Relational Data. Co-advisors: Peter J. Haas, Alexandra Meliou, and Azza Abouzied (NYU)

M.S. in Computer Science
University of Bologna
Bologna, Italy · 2011 – 2013
Summa cum laude. Thesis: Temporal Information Retrieval. Advisor: Danilo Montesi

B.S. in Computer Science
University of Bologna
Bologna, Italy · 2008 – 2011
Summa cum laude. Thesis: Mobile systems for patient-reported outcomes. Advisor: Danilo Montesi

EXPERTISE

- Prescriptive analytics & optimization

SUMMARY

Co-founder and Chief Scientific Officer of OSM Data, where our team is building the first **Decisionhouse** — making prescriptive analytics and decision-making native to the data stack. At its core is **DeQL**, a SQL-like language that turns data into optimal decisions. Previously a researcher at Microsoft Research: ~$100M/yr savings from Azure VM Allocator improvements, 24× SQL Server tuning speedup (patent filed), 10% accuracy lift on LLM-to-optimization translation. PhD from UMass Amherst, pioneering the in-database optimization behind OSM — recognized with a CACM Research Highlight.

EXPERIENCE

Co-Founder & Chief Scientific Officer
Sept 2025 – present
OSM Data · Seattle, WA, USA

- Building the first Decisionhouse — a new category of data system in which decision-making is a first-class primitive alongside storage and analytics (VLDB '26 vision).
- Designing DeQL, a declarative SQL extension for prescriptive queries; shipped DeQL Studio, the interactive interface (VLDB '26 demo).
- Leading our founding team across engineering, research, and design.

Researcher
Oct 2021 – July 2025
Microsoft Research · Redmond, WA, USA
MLO (Machine Learning & Optimization) Group · Data Systems Group

- Azure VM Allocator — projected ~$100M/yr savings from a 1% packing-density gain and CPU-stranding reduction; deployed to production.
- Wred (SIGMOD '24) — 24× SQL Server index-tuning speedup via query-rewriting that minimizes optimizer planning cost while preserving tuning accuracy; patent filed.
- OptiGuide (NL2Opt with LLMs) — 10% formulation-accuracy lift via chain-of-thought prompting, automatic syntax/feasibility repair, and a verifier + disambiguation module; demoed to Azure CVPs; integrated with Fabric's Data Agent.
- Acted as the technical bridge between research and product teams, driving rapid prototyping and iterative deployment across multiple stakeholders.

Research Intern
May – Aug 2017
Microsoft Research · Redmond, WA, USA

- NL-to-SQL using neural networks.

Research Intern, Watson Health
Jun – Aug 2016
IBM Watson · T. J. Watson Research Center, Yorktown Heights, NY, USA

- “Conceptual queries” for healthcare applications.

Research Assistant
Sep 2013 – May 2021
DREAM Lab, University of Massachusetts · Amherst, MA, USA

- Created Package Builder, a system that extends query engines to support the generation of *packages* — collections of tuples with global properties, computed as integer linear programs.
- Technical papers at VLDB 2024, SIGMOD 2020, VLDB 2016; demo papers at VLDB 2020 and VLDB 2014.
- Invited journal papers at CACM Research Highlights 2019, VLDB Journal 2018, SIGMOD Records 2017.

Visiting Ph.D. Candidate
Mar – May 2016
Design Technology Lab, NYU Abu Dhabi · Abu Dhabi, UAE

- Database systems & query optimization
- LLM-powered systems (NL⟶optimization)
- Scalable & fair machine learning

- Paper: Redistributing Funds across Charitable Crowdfunding Campaigns (arXiv:1706.00070).

Visiting Ph.D. Student
May – Aug 2015

AMPLab, UC Berkeley · Berkeley, CA, USA

- Worked on extensions to Package Builder to identify ideal *packages* by example, under an NYU summer internship.

Visiting M.S. Student
Jan – Jun 2013

Data-Intensive Systems Lab, Aarhus University · Aarhus, Denmark

- M.S. thesis on Temporal Information Retrieval under Christian S. Jensen.
- Published two papers in leading NLP and IR venues during the visit.

Undergraduate Research Intern
Jan – Jun 2013

Database Lab, UC Riverside · Riverside, CA, USA

- Worked with Vassilis J. Tsotras on query optimization for query-result diversification.
- Built parser + optimizer + client/server stack for a SQL-like diversification query language.

SELECTED PUBLICATIONS

Decisionhouse: Prescriptive Analytics in the Data Stack

Matteo Brucato, Fjodor Kholodkov, Soren Little, Jakob Mayer, Duc Nguyen

VLDB 2026

DeQL Studio: Declarative Decision-Making over Relational Data

Matteo Brucato, Fjodor Kholodkov, Soren Little, Jakob Mayer, Duc Nguyen

VLDB DEMO 2026

Stochastic SketchRefine: Scaling In-Database Decision-Making under Uncertainty to Millions of Tuples

Riddho Ridwanul Haque, Anh Mai, **Matteo Brucato**, Azza Abouzied, Peter Haas, Alexandra Meliou

VLDB 2025

Wred: Workload Reduction for Scalable Index Tuning

Matteo Brucato, Tarique Siddiqui, Wentao Wu, Vivek Narasayya, Surajit Chaudhuri

SIGMOD 2024

Scaling Package Queries to a Billion Tuples via Hierarchical Partitioning and Customized Optimization

Anh L. Mai, Pengyu Wang, Azza Abouzied, **Matteo Brucato**, Peter J. Haas, Alexandra Meliou

VLDB 2024

sPaQLTooLs: A Stochastic Package Query Interface for Scalable Constrained Optimization

Matteo Brucato, Miro Mannino, Azza Abouzied, Peter J. Haas, Alexandra Meliou

VLDB DEMO 2020

Stochastic Package Queries in Probabilistic Databases

Matteo Brucato, Nishant Yadav, Azza Abouzied, Peter J. Haas, Alexandra Meliou

SIGMOD 2020

Scalable Computation of High-Order Optimization Queries

Matteo Brucato, Azza Abouzied, Alexandra Meliou

CACM 2019

Scalable Package Queries in Relational Database Systems

Matteo Brucato, Juan Felipe Beltran, Azza Abouzied, Alexandra Meliou

VLDB 2016

SELECTED INVITED TALKS

Enabling In-Database Decision Support with Package Queries

Gray Systems Lab, Microsoft · virtual · Mar 2024

Package Queries: Scalable Prescriptive Analytics Close to the Data

Georgia Tech · virtual · Mar 2021

Package Queries: Scalable Prescriptive Analytics Close to the Data

Carnegie Mellon University · virtual · Mar 2021

Package Queries: Enabling Declarative and Scalable Prescriptive Analytics in Relational Data

Harvard University, DASlab · virtual · Jan 2021

Stochastic Package Queries for In-Database Optimization Under Uncertainty

INFORMS 2020 · virtual · Nov 2020

Package Queries: Enabling Declarative and Scalable Prescriptive Analytics in Relational Data

MIT CSAIL, Data Systems Group · Cambridge, MA · Feb 2020

Scalable Package Queries in Relational Database Systems

Microsoft Research · Redmond, WA · Jun 2017

Package Queries: From Tuples to Packages

AMPLab, UC Berkeley · Berkeley, CA · Jun 2015

SERVICE

PROGRAM COMMITTEE ICDE 2027, VLDB 2027, SIGMOD 2026, SIGMOD 2025, VLDB 2025, ICDE 2025, DataPlat workshop @ ICDE 2024, IEEE Transactions of Knowledge and Data Engineering (TKDE) 2020

HONORS & AWARDS

2025 Distinguished Reviewer, SIGMOD 2025 — Berlin, Germany
2020 Best Demo Award, VLDB 2020 (sPaQLTooLs) — Tokyo, Japan
2020 Best Demo Runner-up Award, VLDB 2020 (SuDocu) — Tokyo, Japan
2019 Krithi Ramamritham Computer Science Scholarship — UMass Amherst
2018 CACM Research Highlight Award
2017 ACM SIGMOD Research Highlight Award
2017 Invited journal paper, VLDB Journal — Special Issue on Best Papers of VLDB 2016
2016 Best Papers of VLDB 2016 — New Delhi, India
2013 M.S. with Honors (summa cum laude), University of Bologna
2013 Best Young Researcher Award Nominee, RANLP 2013 — Hissar, Bulgaria
2013 Merit Scholarship — awarded to 0.2% of all students of the University of Bologna
2012 Research Scholarship, University of Bologna
2011 B.S. with Honors (summa cum laude), University of Bologna